

Vectorized functions: a vector can also be a multi-variable function:

• Recall, a real-valued function:

$$f: D \rightarrow \mathbb{R}$$

D : Domain

• Vector-valued function:

$$F: D \rightarrow \mathbb{R}^3$$

$\cap$

$\mathbb{R}$

$$t \mapsto \vec{F}(t)$$

• In general,

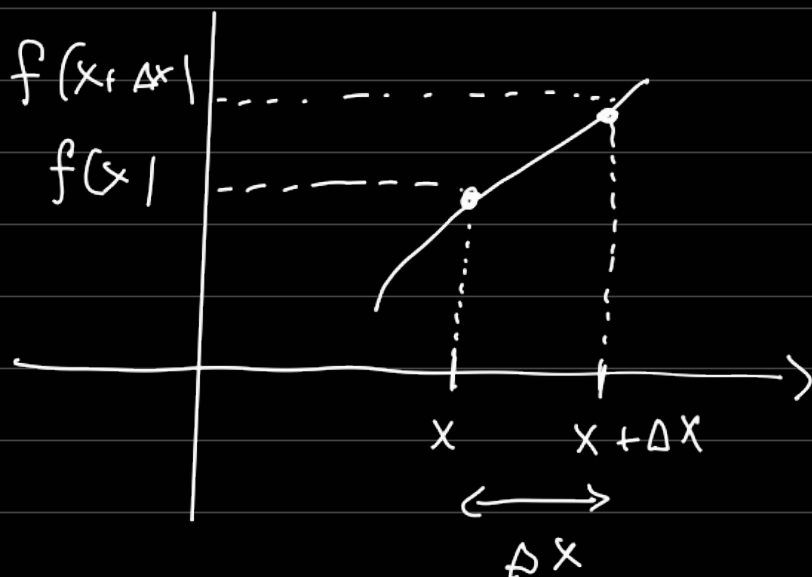
$$\vec{F}(t) = F_x(t)\hat{e}_x + F_y(t)\hat{e}_y + F_z(t)\hat{e}_z$$

• Derivative of a real-valued function:

$$f: D \rightarrow \mathbb{R}$$

then

$$\frac{df}{dx} = f'(x) = \lim_{\Delta h \rightarrow 0} \left(\frac{f(x + \Delta h) - f(x)}{\Delta h} \right)$$



• Also: $x_2 \equiv x_1 + \Delta x$
 $x_1 \equiv x$

$$\Delta x = x_2 - x_1$$

$$\frac{df}{dx} = \lim_{x_1 \rightarrow x_2} \left(\frac{f(x_2) - f(x_1)}{x_2 - x_1} \right)$$

Derivative of a vector-valued function:

$$\frac{d\vec{F}(t)}{dt} = \lim_{\Delta t \rightarrow 0} \left(\frac{\vec{F}(t + \Delta t) - \vec{F}(t)}{\Delta t} \right)$$

Recall: $\vec{F} = F_x \hat{e}_x + F_y \hat{e}_y + F_z \hat{e}_z$

and for $\vec{A} + \vec{B} = (A_x + B_x)\hat{e}_x + (A_y + B_y)\hat{e}_y + \dots$

So

$$\frac{\vec{F}(t + \Delta t) - \vec{F}(t)}{\Delta t} = \frac{F_x(t + \Delta t) - F_x(t)}{\Delta t} \hat{e}_x + \frac{F_y(t + \Delta t) - F_y(t)}{\Delta t} \hat{e}_y + \dots$$

But

$$\frac{F_x(t + \Delta t) - F_x(t)}{\Delta t} = \frac{dF_x(t)}{dt}$$

Thus: $\frac{d\vec{F}}{dt} = \frac{dF_x}{dt} \hat{e}_x + \frac{dF_y}{dt} \hat{e}_y + \frac{dF_z}{dt} \hat{e}_z$

• If $\hat{e}_x, \dots$ are our canonical bases then we can think of $\frac{d}{dt}$ as an operator

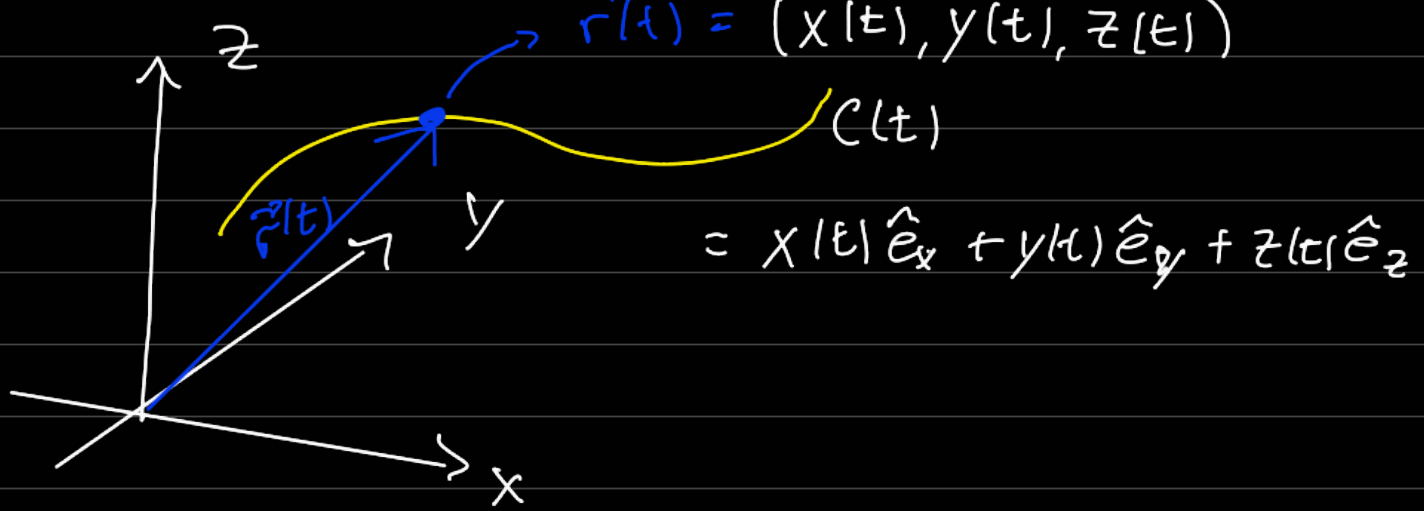
$$\frac{d}{dt}(\vec{F}) = \frac{d}{dt} \left(F_x \hat{e}_x + \dots \right)$$

$$= F'_x \hat{e}_x + F'_y \hat{e}_y + F'_z \hat{e}_z$$

Physics: $x(t)$: position of a particle in the x -direction

$y(t)$:

$z(t)$:



• We say the curve $C(t)$ is parametrized by t .

• Velocity: Rate of change of position

1D | $V_x(t) = \frac{dx(t)}{dt}$

3D | $\vec{V}(t) = \frac{d\vec{r}}{dt} = \dot{x}\hat{e}_x + \dot{y}\hat{e}_y + \dot{z}\hat{e}_z$

Acceleration: $\vec{a}(t) = \frac{d\vec{V}}{dt} = \frac{d^2\vec{r}}{dt^2}$

$$\vec{a}(t) = \ddot{x}\hat{e}_x + \dots$$

Speed: Speed is the magnitude of the velocity vector:

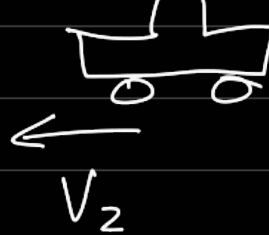
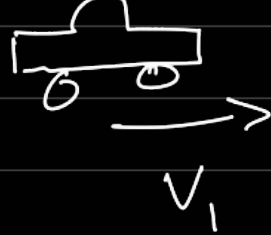
$$|\vec{V}|^2 = \vec{V} \cdot \vec{V} = V_x^2 + V_y^2 + V_z^2$$

$$\frac{d}{dt}(\phi \vec{A}) = \phi' \vec{A} + \phi \vec{A}'$$

* Integrals:

$$\int \vec{A}(t) dt = \int A_x dt + \dots$$

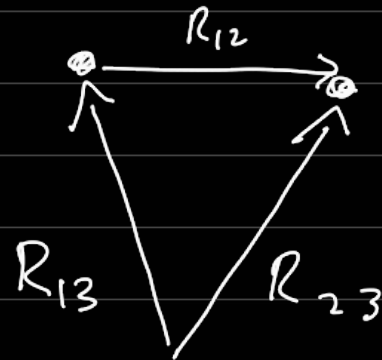
Q41.



$$V_{21} = V_{23} - V_{13} \\ = -V_2 - V_1$$

$$V_{13} = V_1 \\ V_{23} = -V_2$$

$$V_{11} = 0 \\ V_{21} = -V_1 - V_2$$



$$V_{22} = 0 \\ V_{12} = V_1 + V_2$$

$$R_{21} = R_{23} - R_{13} \\ R_{12} = R_{13} - R_{23}$$

Line-integrals:

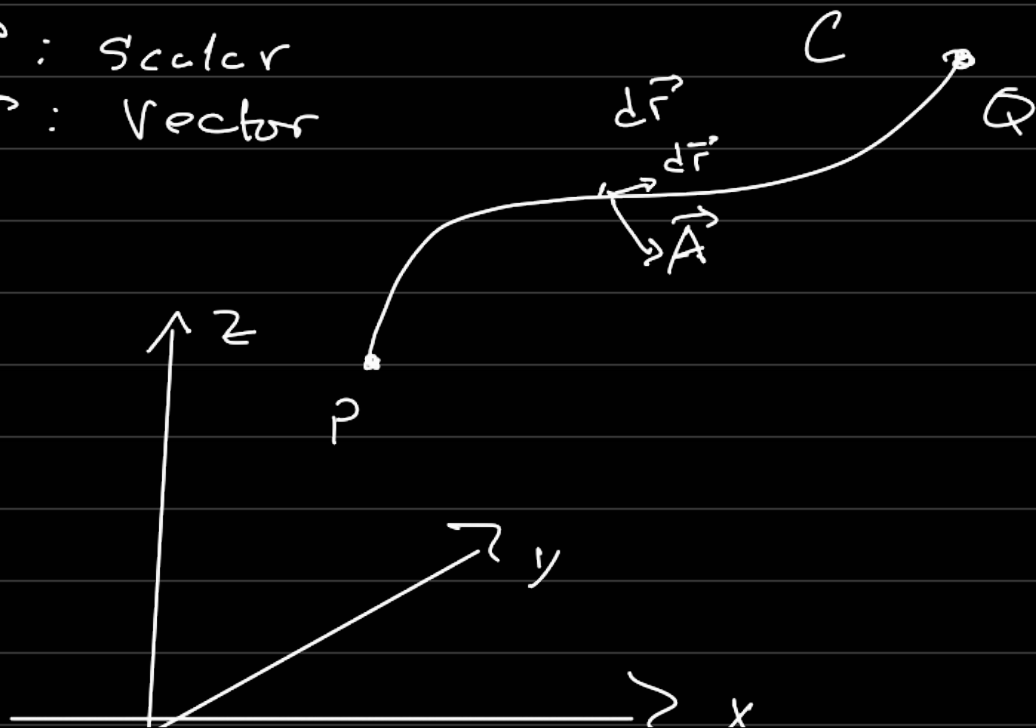
* Integrals along a Curve:

- Has different forms

$$\int \phi d\vec{r} : \text{Vector}$$

$$\int \vec{A} \cdot d\vec{r} : \text{Scalar}$$

$$\int \vec{A} \times d\vec{r} : \text{Vector}$$

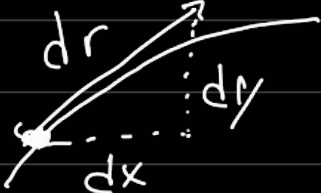


along C we follow via $\vec{r}$

$$d\vec{r} = (dx, dy, dz)$$

$$\int_P^Q \vec{A} \cdot d\vec{r}$$

$(\vec{A} \cdot d\vec{r})$: tangential component of $\vec{A}$ along C

* Arc length: 

$$d\vec{r} = dx \hat{e}_x + dy \hat{e}_y + dz \hat{e}_z$$

$$|d\vec{r}|^2 = d\vec{r} \cdot d\vec{r}$$

$$|d\vec{r}|^2 = dx^2 + dy^2 + dz^2$$
$$|d\vec{r}| = \sqrt{dx^2 + dy^2 + dz^2}$$

$$\int_P^Q \hat{n} \cdot d\vec{r}$$

$$\hat{n} = \frac{d\vec{r}}{|d\vec{r}|}$$

$$L = \int_P^Q \hat{n} \cdot d\vec{r} = \int_{t_0}^t |d\vec{r}| = \int_{t_0}^t \sqrt{dx^2 + dy^2 + dz^2}$$
$$= \int_{t_0}^t \sqrt{(dx^2 + dy^2 + dz^2) \frac{dt^2}{dt^2}}$$

• $ds \equiv |d\vec{r}|$ "line-element"

* Most important "tool" in Einstein's Relativity

